Zero Paradox Companion

Where the First Step Fails

The Real Numbers as Counterexample

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This companion document is written for general readers. It explains in plain language why, on the real number line, 0 has no first step above it, AX-B1's form, and why T-SNAP's shape, the Binary Snap, does not need one; and where the 2-adic numbers Q_2 stand. Density is the reason $\mathbb{R}$ has no first step above 0 (f_snap_impossible). The topological difference from Q_2 is that $\mathbb{R}$ is connected, so not totally disconnected (real_no_snap), while Q_2 is totally disconnected (padic_snaps). That is not a separate fence against a first step: for a linear order carrying its order topology, connectedness implies density (Mathlib denselyOrdered_of_preconnectedSpace). Q_2's topology supplies no first step: Mathlib gives Q_2 no order, and under the norm preorder 0 has no first step there either (ZeroParadox/BottomCannotBe.lean). Neither carrier supplies the first step at $\perp$ of the state carrier, the set of states a run of the framework moves through, ordered as in ZP-A (that first step is AX-B1, a commitment). The formal results are machine-verified in Lean 4 as part of ZP-F (F-SNAP-IMPOSSIBLE, f_snap_impossible: no linearly ordered field has a least positive element, so 0 has no first step above it, AX-B1's form); see also ZP-B (p-adic topology), where Q_2 is totally disconnected (t5_totallyDisconnected, an instance Mathlib already supplies) and that topology supplies no first step.

I. The Natural Question

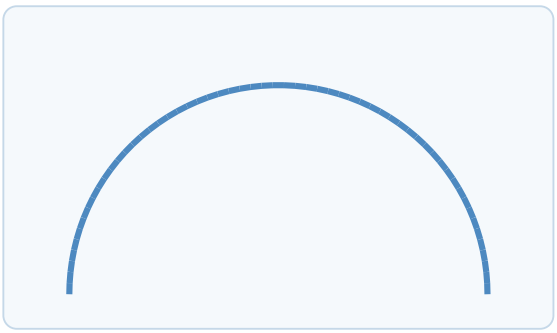
If you have read ZP-B, a natural question arises: why the 2-adic metric? Why p-adic numbers at all? The real numbers are more familiar and more widely used. They seem like the natural substrate for any framework involving continuity, limits, and state transitions. This document is the answer.

The answer begins with a counterexample. The real number line lacks the first step AX-B1 commits the framework's carrier to — not because it is wrong, but because no ordered field, $\mathbb{R}$ and $\mathbb{Q}$ among them, has a least positive element (f_snap_impossible, which, despite its name, states exactly that), so 0 has no first step above it. T-SNAP's shape does not need a first step: t_snap_given holds at a step taken in a densely ordered join-semilattice ($\mathbb{Q}_{\geq 0}$ under max, ZeroParadox/Order/Snap.lean). Understanding why $\mathbb{R}$ has no first step above 0 is the clearest path to seeing what Q_2 removes and what it does not supply.

II. The Density Symmetry

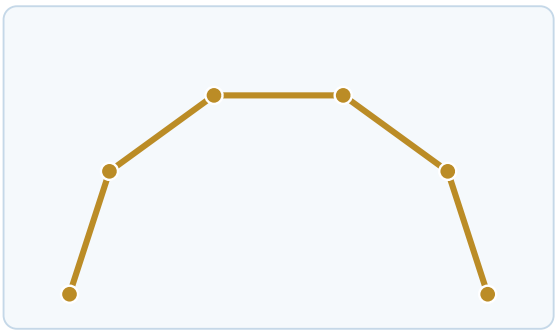
The smooth nature of the real numbers comes from the fact that between any two real numbers — no matter how close — there is always another one. This density applies uniformly: for any proposed first step $\varepsilon > 0$ away from zero, the value $\varepsilon/2$ is smaller and also positive. There is no minimal departure. The density that prevents a closest real number to zero is the same density that prevents a smallest positive real number — it works in every direction.

Zero in $\mathbb{R}$ is not a special topological location. It is an ordinary point on a continuum that looks the same from every direction. For any candidate first step $\varepsilon_0 > 0$, the number $\varepsilon_0 / 2$ also exists, and $\varepsilon_0 / 4$, and $\varepsilon_0 / 2^n$ for every n. So in $\mathbb{R}$ 0 has no first step above it, AX-B1's form (f_snap_impossible: no linearly ordered field has a least positive element). T-SNAP's shape, the Binary Snap, does not need one: t_snap_given holds at a step taken in a densely ordered join-semilattice.



Standard scale — smooth and continuous

No first step away from zero



Zoomed in — the same arc shows discrete steps

A minimum unit of departure becomes visible

The same mathematical arc — a piece of a circle, the curve that defines π — at two sampling resolutions. At standard scale it appears smooth and continuous (left). At high zoom, discrete steps become visible (right). In the actual real numbers, the decimal expansion can always continue; the smooth curve never structurally breaks down. A first step above 0, AX-B1's form, needs a space where that breakdown is built in, not imposed; T-SNAP's shape, a join step off ⊥ of the state semilattice to a different state (t_snap_given), does not, and the cover is a further hypothesis (t_snap_given_cover).

III. Where the Infinity Lives

The distinction between $\mathbb{R}$ and $\mathbb{Q}_2$ is not a matter of degree. It is a question of where the infinity lives.

In $\mathbb{R}$, the infinity lives in the representation. An infinitely long decimal like $\pi = 3.14159\ldots$ describes a finite magnitude — a number sitting between 3 and 4. The infinite expansion is the notation, not the thing being noted. The number itself is finite and well-located on the line. Zero is a limit point: surrounded by non-zero reals, reachable as a limit but not isolated from them. That is a fact about the metric, and the 2-adic zero shares it, so being a limit point does not separate the two. The topology does: $\mathbb{R}$ is connected (place_topological_signature), so not totally disconnected (real_no_snap), and $\mathbb{Q}_2$ is totally disconnected (t5_totallyDisconnected). The order separates them globally, not at zero. At zero neither has a first step: $\mathbb{R}$ in its field order (axb1_fails_in_ordered_field), and $\mathbb{Q}_2$ in the norm preorder of ZeroParadox/BottomCannotBe.lean, § ROLE, where that fact holds in every nontrivially normed field. Globally $\mathbb{R}$ is densely ordered (axb1_fails_everywhere_iff_dense), in its norm preorder as in its field order, while that norm preorder on $\mathbb{Q}_2$ is not dense: 2 has the first step 1 (the § ROLE example stating that cover).

In $\mathbb{Q}_2$, the infinity is the address of zero. The 2-adic valuation assigns $+\infty$ to 0 as its actual value, and no non-zero element shares it (padic_addVal_eq_top_iff); every non-zero element carries a finite integer valuation. That is the valuation chart. In the metric chart the 2-adic zero is a limit too: doubling a non-zero element over and over converges to 0 and does not reach it in finite time, while counting down in the natural numbers reaches 0 in finitely many steps (floor_reach_separates_mu_nu). $\mathbb{Q}_2$'s zero is a floor approached as a limit, not one stepped onto. Neither fact is a first step.

A geometric analogy: the Riemann sphere maps the entire complex plane onto a sphere of diameter 1, placing the origin at one pole and the point at infinity at the other. The two are antipodal — as far apart as any two points on the sphere can be. The 2-adic valuation does something structurally similar: it places zero at infinite valuation and every non-zero element at finite valuation, making zero and the rest of the number

line antipodal in the valuative sense. What the Riemann sphere shows geometrically, the 2-adic valuation encodes algebraically.

$\mathbb{R}$ Real Numbers	$\mathbb{Q}_2$ 2-adic Numbers
Order: densely ordered, in its field order and in its norm preorder alike, so no element, zero included, has a first step	Order: no field order synthesizes in Mathlib. Under the norm preorder 0 is least and has no first step (between 0 and any non-zero element lies one of smaller norm, as in any nontrivially normed field), yet globally it is not dense (2 has the first step 1); under the valuation order 0 is greatest
Topology: connected (place_topological_signature), so not totally disconnected (real_no_snap)	Topology: totally disconnected (t5_totallyDisconnected)
Metric: 0 is a limit point of the non-zero reals	Metric: 0 is a limit point too — the doubling orbit $2^n x$ tends to 0 and does not reach it in finite time
Infinity lives in the representation (infinite decimal = finite magnitude)	Valuation: infinity is the address of 0 — $v_2(0) = +\infty$, the greatest value; every nonzero element has a finite valuation
No first step above 0 (AX-B1's form): density puts a smaller positive element below every candidate (f_snap_impossible: no linearly ordered field has a least positive element). T-SNAP's shape does not need a first step (t_snap_given)	The topological obstruction to an irreversible step is removed (c3_irreversible: no continuous path in $\mathbb{Q}_2$ from a nonzero point to 0; a case of Mathlib's TotallyDisconnectedSpace.eq_of_continuous: in $\mathbb{Q}_2$ no continuous path joins any two distinct points, so the step out from 0 is blocked as well), and no first step at 0 is supplied (AX-B1's first step is a cover of $\perp$ on the state carrier, a commitment)

IV. Finite Precision Supplies a First Step

Every computer already knows this. Any number system with a maximum number of decimal places — any fixed-point arithmetic system, any discretized simulation — has a smallest representable positive number — call it δ. Below that, there's nowhere to go. The density argument fails at δ: there is a genuine first step above 0, AX-B1's form, and halving it is not possible. T-SNAP's shape is a join step off $\perp$ of the state semilattice to a different state (t_snap_given) and does not need it; that a run's first transition is a cover is a further hypothesis (t_snap_given_cover). Whether a step is taken is a separate question, and the framework commits to it rather than deriving it.

The real numbers are the idealization in which this floor is removed — the limiting case in which precision is unbounded and the minimum disappears. That is not a feature from the perspective of the Zero Paradox. It is exactly what rules out a first step above 0.

$\mathbb{Q}_2$ is not an artificially truncated real line, and it does not restore a smallest step. Its zero is a floor of the other kind: a finite-precision system's floor is reached by counting down, while $\mathbb{Q}_2$'s is a limit the doubling orbit approaches and does not reach in finite time (floor_reach_separates_mu_nu), marked out by its valuation, $v_2(0) = +\infty$. The first step AX-B1 commits to is a cover of $\perp$ on the state carrier (a commitment): a

state directly above ⊥ with nothing strictly between, which order theory calls an atom. t_snap_given_cover assumes the run's first transition is a cover; with CC-1, the commitment that the run starts at ⊥, that transition is a cover of ⊥. MachinePhase meets the commitment (its state c_1 covers its ⊥, c_0). In neither ordering of Q_2 does its 0 have a first step, so the step is not a consequence of the 2-adic structure.

> Every finite-precision number system with a positive representable value already has a first step above 0, AX-B1's form: its least positive value. The minimum representable positive value exists; the density argument fails there. The real numbers are the most familiar example of a structure that removes this floor — but any dense ordered set has the same property, including the rational numbers: for any $\varepsilon_0 > 0$, $\varepsilon_0 / 2$ also exists. Q_2 does not put a smallest step back in: its zero is a floor approached as a limit and marked by its valuation, and the step itself is AX-B1.

V. The Coach and the Players

The real number line is not wrong. It is internally consistent and extraordinarily useful for calculus, analysis, and modeling continuous change. But on the line, 0 has no first step above it, AX-B1's form — not because the line fails as a mathematical structure, but because of how it treats zero.

Consider a sports team. The coach and the players are not peers. The coach is not on the field, does not wear the same uniform, and cannot be substituted in. The role is categorically different — the coach is the origin and organising principle of the game, not a participant in it. You cannot return to the coach the way you move between players. That path does not exist.

Zero has the same relationship to the states above it. It is the floor from which everything departs — not a peer of the states, but their origin. In the ZP framework, zero is the asymptotic limit that the system moves away from, not a state it can stably occupy or return to. In ℝ, zero has no such character: it is a regular point of a dense line, indistinguishable in structure from 1 or π, with no first step above it. Density is what disqualifies it (axb1_fails_everywhere_iff_dense). The line is also connected (place_topological_signature), so not totally disconnected (real_no_snap), which is its topological difference from Q_2; for a line carrying its order topology, connectedness implies density (Mathlib denselyOrdered_of_preconnectedSpace), so that is not a second reason.

The real number line treats zero as an ordinary point — a peer of every other element, with the same local structure as 1 or π. In ℝ, zero is not valuatively distinct from any other element; every point looks the same from the perspective of the metric. That is why return paths are permitted: if zero is just another player, nothing distinguishes the path back from any other movement on the line. Directionality — the one-way ratchet the framework requires — cannot be grounded in a space where zero is indistinguishable in kind from everything else.

Q_2 gives zero a genuinely different role, and it shows in two charts. In the valuation chart zero has the address $+\infty$: $v_2(0) = +\infty$, the greatest value, while every non-zero element carries a finite integer valuation, and only zero has $+\infty$ (padic_addVal_eq_top_iff). In the metric chart zero is a limit the other elements approach: the doubling orbit converges to it and does not arrive in finite time (floor_reach_separates_mu_nu). Zero is in Q_2 but it is not a peer. The coach is not on the field. The path back does not exist.

VI. Mathematical Constants and Forcing

A natural follow-on question: if infinitely long decimals are the issue, what about π itself? π is infinitely long. Does it already qualify for the incompressibility results elsewhere in the ZP framework?

No — and the reason matters. π is infinitely long in its decimal expansion, but it is computable. Finite algorithms — the Leibniz formula, the BBP algorithm, dozens of others — can generate π to any precision you specify, however large. There is no bound on the precision you can ask for. But this is the point, not a problem: the Kolmogorov complexity of the first n digits of π is tiny relative to n. The short algorithm is the information, not the infinite decimal. You can demand arbitrarily many digits; the specification of π remains short regardless.

This is what makes π a constant rather than an arbitrary number. It is the necessary consequence of a geometric relationship — the ratio of circumference to diameter — and that relationship provides the compression. The value is forced by the definition. The computability follows from the forcing. Constants are computable because they have finite definitions; the finite definition is the short program.

Most real numbers are not like this. Almost all real numbers — in the measure-theoretic sense — have no finite definition that picks them out. They are algorithmically random: no short program generates them; the first n digits have Kolmogorov complexity close to n. High algorithmic complexity is the signature of a number with no mathematical structure behind it, no definition that points to it specifically.

VII. Two Kinds of Incompressibility

This brings us to a subtle but important point. ZP-C includes the result L-INF: $\perp$ has unbounded surprisal — no finite external description can capture it. A careful reader might ask: is this the same as saying $\perp$ is algorithmically incompressible, like a random real number?

It is not. The distinction is the difference between randomness and necessity.

A random real number is incompressible because it has no structure — no mathematical relationship forces it to be what it is. Its complexity is high because it is arbitrary. Nothing points to it specifically.

$\perp$ is incompressible for the opposite reason: because anything standing external to it is structurally excluded. Nothing can occupy a position outside $\perp$ to describe it — not because the description is too complex, but because the position of "external to $\perp$" does not exist. $\perp$ is not arbitrary. It is the most constrained object in the framework, the unique global minimum of the lattice. Its incompressibility is not the noise of randomness. It is the silence of structural necessity.

The real number line mixes both kinds — computable constants alongside algorithmically random reals — with no topological distinction between them. ZP-C's L-INF is a claim of an entirely different character.

The contrast in one sentence

A random real is incompressible because nothing forced it to be what it is. $\perp$ is incompressible because nothing can stand outside it. One is the absence of structure. The other is structure all the way down.

VIII. Two Limit Conditions, Side by Side

The density argument and the ordinal threshold argument are two separate observations — one about why 0 has no first step above it in $\mathbb{R}$, one about how ε_0 of Ordinal is reached. Read side by side they share a shape, and the shared shape is an analogy, not a proved identification.

From the field side: density shows there is no smallest positive element in $\mathbb{R}$. For any candidate first step $\varepsilon > 0$, the value $\varepsilon/2$ is smaller and also positive. There is always something between any candidate step and zero, so zero has no first step. That is a fact about the order. In fact an order is dense exactly when no element at all has a first step (axb1_fails_everywhere_iff_dense), so density rules out a first step at zero; one element lacking a first step would not make an order dense. Zero in $\mathbb{R}$ is also a limit point, a fact about the metric, and so is zero in Q_2 (floor_reach_separates_mu_nu).

From the ordinal side: ε_0 of Ordinal has no ordinal immediately below it. There is no α with $\alpha + 1 = \varepsilon_0$. It is the limit of the tower ω, ω^ω, ω^ω^ω, … — approachable from below by finite iteration, but not reachable in any finite number of steps.

The shared shape: in the field case, zero has no smallest positive neighbour in $\mathbb{R}$; in the ordinal case, ε_0 has no immediate predecessor in the ordinals. Each is a point approached by a limit and not reached in finitely many steps from one side. These are two limit-type conditions in two different structures, not one boundary. What is proved (cnf_bridge_type_boundary) pairs a face of ε_0 other than that limit one, its least-fixed-point face, with $\mathbb{Z}_2$, the 2-adic integers inside Q_2, and as co-witnesses, not as an identity: ε_0 is the least fixed point of $\alpha \mapsto \omega^\alpha$ in the ordinals, and the 2-adic images (cnfToZp2) of the same tower's codes, from its first stages 0, 1, ω, ω^ω, …, converge to the 2-adic 0. ε_0 is at once that least fixed point and the supremum of those stages (epsilon0_min_eq_max). Read through the 2-adic norm, the same convergence is the 2-adic norms of the images tending to 0 in $\mathbb{R}$ (the norm-chart example in ZeroParadox/Ordinal/CnfBridge.lean § VI); there $\mathbb{R}$ holds the norm values, a different role from $\mathbb{R}$ in the density argument. An equation between ε_0 and the 2-adic 0 does not type-check.

Neither condition is needed for the snap itself. In the two-state carrier MachinePhase, T-SNAP holds (t_snap_derived), and the landing c_1 covers c_0, the bottom of MachinePhase (the MachinePhase example in ZeroParadox/Order/Snap.lean stating that c_1 covers $\bot$ of MachinePhase), so the landing has an immediate predecessor and no limit is involved on either side. What the density argument shows is narrower: a dense order has no first step at any element (axb1_fails_everywhere_iff_dense), so it cannot meet AX-B1's commitment.

What the Counterexample Shows

The real numbers are the most natural, most familiar metric space in mathematics. They are also a space where 0 has no first step above it, AX-B1's form (f_snap_impossible: no linearly ordered field has a least positive element), while T-SNAP's shape does not need one (t_snap_given holds at a step taken in a densely ordered join-semilattice). They are densely ordered, so no element, zero included, has a first step: every candidate first step admits an infinite sequence of smaller steps below it. Zero is also a limit point there, but so is the 2-adic zero, so being a limit point does not separate them. The order separates them globally, not at zero: at zero neither has a first step, and globally $\mathbb{R}$ is densely ordered (axb1_fails_everywhere_iff_dense), in its norm preorder as in its field order, while the norm preorder on Q_2 is not; the norm-preorder facts are the § ROLE examples in ZeroParadox/BottomCannotBe.lean. In the topology $\mathbb{R}$ is connected (place_topological_signature), so not totally disconnected (real_no_snap), and Q_2 is totally disconnected (t5_totallyDisconnected); that difference separates them too. It is not a separate fence against a first step: for

a linear order carrying its order topology, connectedness implies density (Mathlib denselyOrdered_of_preconnectedSpace), so it adds no reason beyond density.

$\mathbb{Q}_2$ is not an exotic choice. Among all completions of the rationals, Ostrowski's theorem says there are exactly two kinds: Archimedean ones (like $\mathbb{R}$, which is densely ordered, so zero has no first step) and non-Archimedean ones ($\mathbb{Q}_p$, where the p-adic valuation assigns zero infinite valuation while every non-zero element has finite valuation). $\mathbb{Q}_2$ is the non-Archimedean completion at $p = 2$. ZP-B's Theorem T0 fixes that prime from binary existence (AX-B1) together with a minimality principle (MP-1), the design choice to use the smallest base that encodes the two states. That zero's valuation is $+\infty$ is set by definition, not produced by the completion; what holds in $\mathbb{Q}_2$ is that zero alone has it (padic_addVal_eq_top_iff). The framework did not choose unusual mathematics for its own sake. ZP-F shows that ordered fields are excluded, by density; ZP-B records that $\mathbb{Q}_2$ is totally disconnected (t5_totallyDisconnected, an instance Mathlib already supplies), which removes the topological obstruction to an irreversible step (c3_irreversible: no continuous path in $\mathbb{Q}_2$ from a nonzero point to 0). Neither supplies the first step at $\bot$ of the state carrier; that is AX-B1.

A note on scope: ZP-F establishes this result for linearly ordered fields — the class that contains $\mathbb{R}$ and $\mathbb{Q}$ — because that is the natural comparison class for the most familiar number systems. The blocking phenomenon is not unique to fields: in any densely ordered structure no element has a first step (axb1_fails_everywhere_iff_dense). The ordered field result is the right frame for the $\mathbb{R}$ comparison. The broader phenomenon is the same.

Where the First Step Fails

In $\mathbb{R}$, 0 has no first step above it, AX-B1's form (f_snap_impossible: no linearly ordered field has a least positive element): for any $\varepsilon_0 > 0$, here any positive real and not the ordinal, $\varepsilon_0 / 2$ also exists; $\mathbb{R}$ is densely ordered, and in a dense order no element has a first step. T-SNAP's shape does not need one: t_snap_given holds at a step taken in a densely ordered join-semilattice. At zero $\mathbb{Q}_2$ has no first step either, in its norm preorder; globally that preorder is not dense, and $\mathbb{Q}_2$ is totally disconnected where $\mathbb{R}$ is connected. Its zero is distinguished in two charts: by its valuation, $v_2(0) = +\infty$ while every non-zero element carries a finite valuation, and as a limit the doubling orbit approaches and does not reach in finite time. Neither supplies the first step AX-B1 commits to, a cover of $\bot$ on the state carrier; t_snap_given_cover assumes the run's first transition is a cover, and with CC-1 (the run starts at $\bot$) that transition is a cover of $\bot$. MachinePhase meets the commitment (c_1 covers its $\bot$, c_0).